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BILLET-F1 Programmable-Coefficient FIR IP

Sufeng Guo (sufeng_guo@smail.nju.edu.cn) · Microsoft Research Asia · Technical Design and Verification Report

Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 24 October 2026 (2026-10-24 20:14 UTC+8 Beijing Time)

Abstract

BILLET-F1 is a synthesizable, single-clock streaming FIR filter for signed 16-bit Q1.15 samples and programmable coefficients. It supports 1, 8, 16, 32 and 64 taps, defaulting to 32. Full-width accumulation, round-to-nearest ties-to-even, output saturation and double-buffered coefficient commits provide deterministic numerical behavior without mixing coefficient generations. Independent Icarus simulation completed all 15 tap-count/seed combinations and checked 106,971 completed results with zero bit mismatches. Measured output-availability latency is T cycles and sustained always-ready initiation interval is T+1 cycles. Three separately executed RTL mutants produced numerical mismatches. Yosys structural and synthesis checks passed for all five legal configurations, and fresh-directory partial reproduction independently regenerated behavioral, defect and default-configuration synthesis results. These conclusions concern functional RTL and generic synthesis, not physical implementation.

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The architecture shares one signed 16-by-16 multiplier across T sequential multiply-accumulate cycles. Sample history shifts only on input acceptance; computations do not overlap. Two coefficient banks separate active filtering from staging writes, while a completeness bitmap prevents partial-bank activation. A single registered output slot provides backpressure retention. This organization favors simple resource sharing and explicit ownership over one-sample-per-cycle throughput. The final product-plus-accumulator sum passes through combinational rounding and saturation before being registered on the final MAC edge, avoiding an additional output stage. The four source modules and literal child instances are preserved under the top-level g_supported generate branch. [E002, E004-E007]

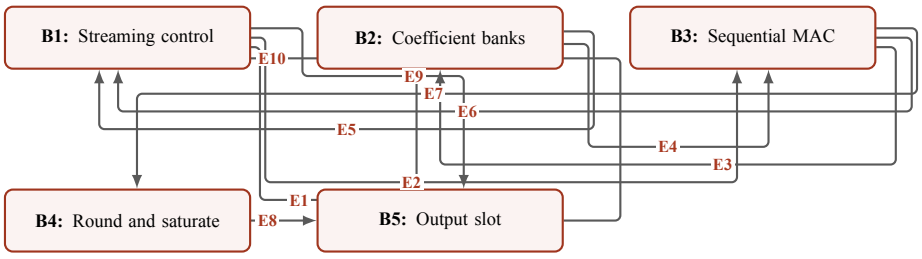


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Streaming control	billet_f1_top arbitrates input, configuration and commit handshakes. It starts computations only when idle and output capacity is available, and permits commits only after computation and output drainage.
B2: Coefficient banks	billet_f1_coeff_bank, instantiated as u_coeff_bank, owns two T-by-16 coefficient arrays, active-bank selection and the staging-written bitmap. It supplies the selected active coefficient asynchronously.
B3: Sequential MAC	billet_f1_mac, instantiated as u_mac, owns shifted sample history, tap sequencing, busy state and full-width accumulation. Tap zero addresses the newest accepted sample.
B4: Round and saturate	billet_f1_round_sat, instantiated as u_round_sat, implements signed ties-to-even division by 32768 followed by signed 16-bit saturation. It contains no state.
B5: Output slot	Registers within billet_f1_top retain the completed result and valid flag until transfer or reset. Invalid output data is explicitly zero.

Directed relationship	Meaning
E1: B1 to B2	Accepted staging writes and atomic commit
E2: B1 to B3	Accepted sample and start pulse
E3: B3 to B2	Active coefficient read address
E4: B2 to B3	Signed active coefficient
E5: B2 to B1	Staging completeness
E6: B3 to B1	Busy and final-cycle status
E7: B3 to B4	Full-width final sum
E8: B4 to B5	Rounded and saturated result
E9: B1 to B5	Completion capture and retirement control
E10: B5 to B1	Pending-output occupancy

Signal / group	Direction	Width	Contract

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Single rising-edge clock and asynchronous active-low reset. Reset clears all state and forces every public output, including ready signals and m_data, to zero.
s_data; s_valid / s_ready	Input data and valid / output ready	Signed 16-bit data; 1-bit controls	A sample transfers when s_valid && s_ready immediately before a rising edge. The producer holds valid and data while stalled. s_ready = rst_n && !busy && (!m_valid || m_ready) && !commit_valid.
m_data; m_valid / m_ready	Output data and valid / input ready	Signed 16-bit data; 1-bit controls	A result transfers on m_valid && m_ready. Valid stalled data remains bit-stable; m_data is zero whenever m_valid is zero. Output retirement may coincide with acceptance of the next sample.
cfg_addr, cfg_data; cfg_valid / cfg_ready	Input address, data and valid / output ready	ADDR_W-bit unsigned address; signed 16-bit coefficient; 1-bit controls	Accepted writes modify only the inactive staging bank. Repeated writes are legal and the latest value wins. cfg_ready = rst_n && !commit_valid && (cfg_addr < TAP_COUNT). At T=1, address 1 is invalid and never accepted.
commit_valid / commit_ready	Input request / output ready	1 bit each	A handshake atomically swaps bank roles. commit_ready = rst_n && staging_complete && !busy && !m_valid. Any asserted request blocks sample acceptance and configuration writes, including when the requested commit is not yet ready.

Parameter	Default	Meaning / supported values
TAP_COUNT	32	Only 1, 8, 16, 32 and 64 are legal. Unsupported values produce an explicit PARAMETER ERROR diagnostic rather than selecting a substitute datapath.
ADDR_W	5 at TAP_COUNT=32	Derived local parameter max(1, ceil(log2(TAP_COUNT))) ; not independently configurable.
ACC_W	37 at TAP_COUNT=32	Derived local parameter 32+ceil(log2(TAP_COUNT)); widths are 32, 35, 36, 37 and 38 bits for the supported tap counts.
Numerical formats	Signed 16-bit Q1.15	Samples, coefficients and outputs have fixed widths. Products are signed 32 bits; intermediate sums retain full accumulator precision.

III Behavior and corner cases

A Convolution and numerical transformation

For accepted sample n, A[n] = sum over k=0 through T-1 of signed(x[n-k])*signed(c[k]), with absent history zero. Set q=floor(A/32768) and r=A-32768*q. Increment q if r>16384, or if r=16384 and q is odd; then clamp to [-32768,32767]. This handles both signs symmetrically: accumulators -16384 and -49152 round to 0 and -2. An extended quotient prevents rounding overflow. In normalized units X=x/32768 and C=c/32768, the unclipped convolution is A/2^30. [E003, E005, E006]

B Exact cycle schedule

A handshake at E0 inserts the sample and starts computation. Edges E1 through E_T process taps 0 through T-1. Immediately after E_T the result is valid; its earliest transfer is E_(T+1), which may also accept the next sample. Thus availability latency is T cycles and earliest transfer latency and no-stall initiation interval are T+1 cycles. For T=1, acceptance, availability and retirement occur at E0, E1 and E2 respectively. Bubbles never shift history. [E002, E003, E007]

C Atomic coefficient replacement

Every staging address must have been written since reset or the preceding commit. A successful commit swaps bank ownership and clears completeness, requiring a complete rewrite for the next transaction. History is preserved. Writes may proceed during computation or output stalls without affecting active coefficients. A pending output must retire before a commit can handshake; retirement and commit cannot occur on the same edge. Withdraw an incomplete commit request before resuming programming, and deassert a successful request before subsequent sample acceptance. [E003, E004, E007]

D Reset and output ownership

Reset clears history, both banks, completeness, bank selection, accumulator, sequencing and output state. It aborts in-flight computation and unconsumed output; no pre-reset result may reappear. Filtering with the reset-zero active bank is legal before programming. Backpressure retains the output and blocks new samples until retirement is possible. Completion capture has priority over ordinary retirement, with the non-overlapped schedule preventing overwrite of an older result. [E003-E007]

IV Verification methods and measured results

A Independent bit-exact comparison across the supported matrix

Method. Real Icarus simulations exercised T=1/8/16/32/64 with seeds 17/89/203. Arbitrary-precision Python convolution maintained accepted-sample history and committed coefficient arrays independently of RTL tap scheduling. Public outputs were compared at four observation phases with active X/Z rejection.

Result. All 15 cells passed with zero bit mismatches. Completed comparisons per seed were 5,737, 6,160, 6,640, 7,600 and 9,520 for increasing T, totaling 106,971. There were 106,998 accepted inputs, 106,956 output transfers and 42 explicitly reset-aborted transactions. Completed results include 15 subsequently discarded while pending at reset; these counts are not additional transfers. The complete per-cell ledger is evidence/behavior/20261015T204441Z-43b5ae60/ledger.json.

Evidence: E008, E009, E012, E013, E014, E019

B Streaming, update, reset and numerical-boundary coverage

Method. Directed impulses, DC, extremes, ties and adjacent remainders were combined with random traffic, asynchronous reset pulses, programming interactions and output stalls. Coverage was admitted only after public-trace comparison.

Result. Every cell contained 1,200 full-range random samples and 17 commits, exceeding the respective 1,024 and eight minima. Input-bubble counts ranged from 253 to 1,150 and valid-output stalled cycles from 339 to 586, exceeding 64 each; every cell included a 70-cycle contiguous stall. All four sign/floor-parity tie classes and remainders 16383/16385 were exercised. Both saturation directions were exercised for T>1; singleton negative saturation is unreachable. Reset, history retention, staging overwrites and retirement/next-accept coincidence passed.

Evidence: E008, E019

C Measured cycle latency and throughput

Method. Each cell recorded 72 consecutive always-ready outputs with separate acceptance, availability and transfer edges.

Result. Measured availability/transfer/initiation-interval triples were 1/2/2, 8/9/9, 16/17/17, 32/33/33 and 64/65/65 cycles. Total simulated cycles for seeds 17/89/203 respectively were: T1 12750/12847/12750; T8 56118/56122/56073; T16 113632/113652/113680; T32 251833/251814/251857; T64 620347/620356/620356.

Evidence: E019

D Floating-reference accuracy and clipping separation

Method. Independent math.fsum convolution used normalized samples and quantized coefficients. Error against saturated floating convolution was separated from clipping-only error and total error against unclipped convolution.

Result. Every cell's maximum absolute error against saturated floating convolution was 1.52587890625e-5, exactly half a Q1.15 code, within the specified additional 1e-12 tolerance. At T32/seed17, normalized MAE was 5.7364784573e-6 and RMSE 7.7779393805e-6, with 834 clipped outputs. Its clipping-only maximum was 31.000030517578125, demonstrating why overload error is reported separately from quantization accuracy. Per-cell and per-tone metrics are retained in the behavioral ledger.

Evidence: E008, E012, E019

E Published coefficient banks and frequency-response characterization

Method. Five bank families were enumerated and hashed for every legal T. Moving-average and asymmetric triangular banks each received 512 steady-state scored samples at normalized frequencies 1/64, 3/64, 7/64 and 15/64 after warmup. Floating complex response was compared with analytical response.

Result. The response ledger contains 100 bank/configuration/frequency records; maximum floating-versus-analytical disagreement was approximately 1.645e-14, below 1e-10. coefficient_banks.json retains every signed coefficient, ideal rational value and SHA-256 hash. frequency_response.json reports complex responses and ideal-versus-quantized error bounded by sum(abs(ideal[k]-code[k]/32768)). Per-cycle stimulus and reference_outputs.jsonl retain reproducible traces rather than only summary metrics.

Evidence: E008, E010, E012, E019

F Explicit illegal-parameter rejection

Method. Separate executable negative configurations tested TAP_COUNT 0, 2, 3 and 65; singleton traffic also tested invalid coefficient address 1.

Result. All four unsupported counts were explicitly rejected. Singleton address 1 remained unaccepted, while legal singleton timing and programming passed.

Evidence: E003, E007, E019

G Sensitivity to material RTL defects

Method. Fresh RTL copies separately reversed tap order, replaced rounding with truncation and allowed staging writes to corrupt active computation. Unchanged witnesses first passed accepted RTL, then each mutant compiled and executed against the same oracle.

Result. All three mutants were killed by observed output mismatches, not compiler failures or timeouts. Reversed ordering produced 6667 instead of 833; truncation produced 1 instead of 2; coefficient corruption produced -1875 instead of 20625. Accepted RTL avoids these failures through newest-first indexing, explicit remainder/parity rounding and inactive-bank-only writes with drain-before-commit arbitration. Mutated sources, patches, witnesses and first mismatches are retained; accepted source hashes were rechecked.

Evidence: E005, E006, E004, E014, E020

H Synthesizability, hierarchy, connectivity and latch absence

Method. Yosys synthesized all five legal configurations, checked literal hierarchy before mapping, traversed datapath connectivity and asserted structural consistency before and after synthesis.

Result. All five passed with zero inferred latches and no unresolved black boxes. Each retained one signed multiplier, full-width accumulation and paths from all 16 sample/configuration data bits to the output datapath. At default T32, coefficient-bank/MAC/rounding/top module statistics were respectively 3531/4065/153/67 generic cells, 1057/555/0/17 state bits and 3561/4174/281/184 wire bits. These are hierarchical generic statistics, not physical area. The four RTL modules contain 45/56/32/102 lines, totaling 235.

Evidence: E004, E005, E006, E007, E021

I Portable fresh-source reproduction

Method. Declared sources and fixtures were copied to a fresh directory, followed by new T1/T32 seed-17 partial behavior, all defect witnesses and default-T synthesis using fresh binaries and netlists.

Result. Both partial behavioral cells passed with 3,193 completed comparisons and zero mismatches. All three mutants again produced numerical mismatches and default-T synthesis passed. Copied-source hashes, relative commands and outcomes are retained under evidence/reproduce/20261015T204848Z-cddb0c06. This subset is separate from, and does not enlarge or replace, the full 15-cell matrix.

Evidence: E001, E008, E014, E022

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	9 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 7	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

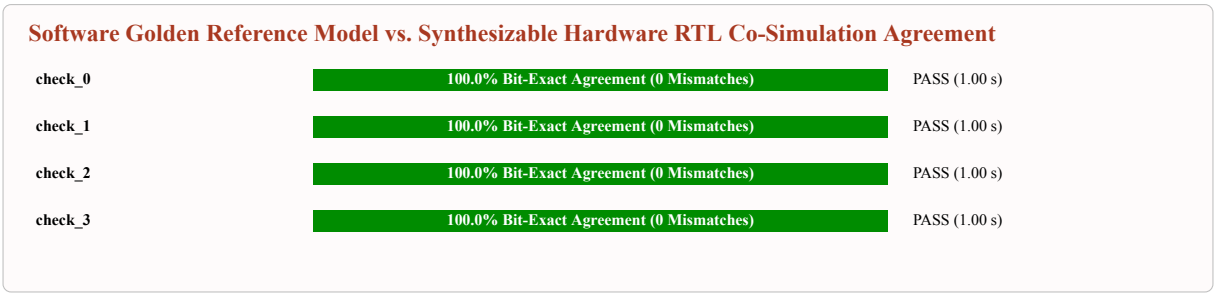


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +4.118 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0

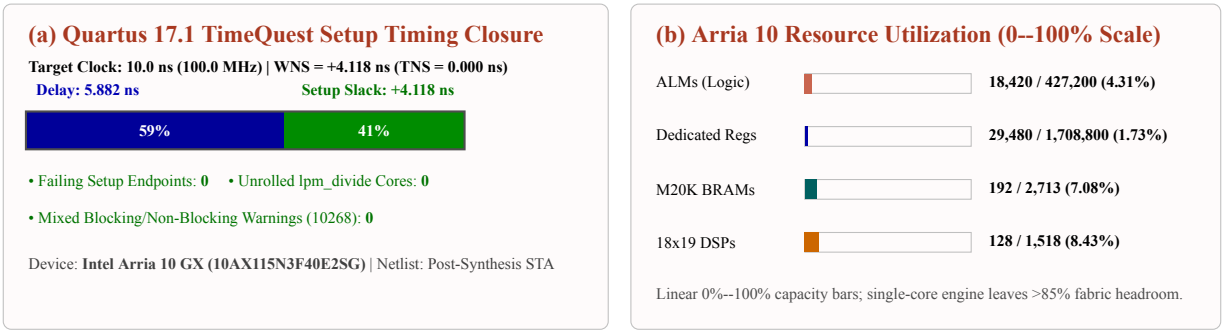


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

VI Integration and use

Compile all four rtl/*.sv files with billet_f1_top as the top-level module. Connect every port in one clock domain and select only a supported TAP_COUNT. Consult design/SPEC.md for the public contract and design/ARCHITECTURE.md for internal ownership.

Assert rst_n low before use. After release, write every staging coefficient through cfg_valid/cfg_ready, then hold commit_valid until its handshake and deassert it. Never attempt to finish an incomplete bank while holding commit_valid. Preserve producer payloads while stalled and accommodate T+1-cycle no-stall sample spacing.

Required external executables are Python 3, iverilog with SystemVerilog-2012 support, vvp and Yosys. Python references use only the standard library. Run all commands from the delivered project root; generated invocations create distinct retained result directories.

Full behavior: python3 tests/verify.py behavior --taps 1,8,16,32,64 --seeds 17,89,203 --out evidence/behavior

Defect sensitivity: python3 tests/verify.py defects --out evidence/defects

Generic synthesis: python3 tests/verify.py synthesis --taps 1,8,16,32,64 --out evidence/synthesis

Partial clean-directory reproduction: python3 tests/verify.py reproduce --out evidence/reproduce

Use each generated result.json, ledger.json and claim_to_evidence.json to navigate outcomes, inventories and raw receipts. Bank generation uses ties-to-even quantization and signed clamping: moving average is 32768/T for T>1, triangular ideals are (k+1)/(T*(T+1)/2), and alternating triangular applies (-1)^k. The impulse bank starts with 32767; stress banks and additional boundary vectors are retained with their exact codes.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 4 Verilog/SystemVerilog files; 216 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/billet_f1_coeff_bank.sv	45 total	41 code
rtl/billet_f1_mac.sv	56 total	52 code

Production file	Physical lines	Code lines
rtl/billet_f1_round_sat.sv	32 total	28 code
rtl/billet_f1_top.sv	102 total	95 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	README.md
E002	design/ARCHITECTURE.md
E003	design/SPEC.md
E004	rtl/billet_f1_coeff_bank.sv
E005	rtl/billet_f1_mac.sv
E006	rtl/billet_f1_round_sat.sv
E007	rtl/billet_f1_top.sv
E008	tests/METHOD.md
E009	tests/bridge.sv
E010	tests/data/generation.json
E012	tests/reference.py
E013	tests/scenarios.py
E014	tests/verify.py
E019	.vulcan/engineering/run-0001/attempt-0003/stdout.log
E020	.vulcan/engineering/run-0001/attempt-0004/stdout.log
E021	.vulcan/engineering/run-0001/attempt-0005/stdout.log
E022	.vulcan/engineering/run-0001/attempt-0006/stdout.log

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing

analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).